1 Scope

The question is whether every threat model in the multi-party learning literature can be written with three objects: data, a model space, and a learning function from a model, a dataset and a hypothesis class to a model. The adversary is then fixed by what it sees, what it chooses, and one of three goals, with no named attack families. The method is to define the interface, map the papers placed on 2026-10-16 onto it, and instantiate it on HE-OFT (KSK26) and on FedAvg, in the terms and symbols of HE-OFT.

2 Result

On three primitives, data, a model space and a learning function that maps a model, a dataset and a hypothesis class to a model, every adversary the sweeps report is a triple: a view of node values, the node arguments and outputs it sets, and a goal that is one of distinguish, invert or steer. Membership inference is a discriminator, reconstruction and model extraction are function inversion, and poisoning and backdoors are steering. Proven: the strength order between two adversaries is post-processing of views together with inclusion of choices, so black-box against white-box is a question of sufficient statistics. HE-OFT already proves one ordering in this language, that its label view is no stronger than holding the shared head, and it proves that a malicious server must be excluded. What HE-OFT leaves unclaimed is steering by deviating clients, and the check that would let it admit a deviating server.

Does not show: The interface is a definition, and definitions in this area are commodities, so its worth lies only in the collapse and separation results it organizes. Whether its novelty survives is not checked. Salem et al., IEEE S&P 2023, unify inference attacks as privacy games (from memory, not placed), and the distinguish and invert half may already be theirs. Every note except HE-OFT's is built from a sweep extraction, so no other row is cited from a full reading.

3 How it was produced

Sources	Four sweeps, venue lists of IEEE S&P, USENIX Security, CCS, NDSS, PoPETs, NeurIPS, ICML, ICLR, CRYPTO, EUROCRYPT. DBLP was rate limited. HE-OFT read in full from `HE-IFD/docs/paper`, commit 9d894ea.
Range	2021 to 2026, anchors before 2024 only where they fixed a threat model
Screened	44 papers placed with first-page author checks
Artifacts	`lib/literature/catalog.md`, `lib/literature/notes/KSK26.md`, `My Drive/HE-ML/pdfs/`
Reproduce	`lib/literature/positioning.md`

4 What this changes

A research problem in this language is one step between two adversaries that the field crosses without a result, with a collapse or a separation proven at a fixed goal and a test suite that measures the gap. For the group, the semi-honest to malicious server step is closed by HE-OFT's own impossibility, so the open object is the cheapest check on what the quorum switches that restores the semi-honest view, which is the question of the verifiable HE study. Küpçü's black-box question asks for the reverse of HE-OFT's ordering, when Q label queries reach what the shared head reveals, and this is open.

5 Method

The three primitives

The interface has three primitives, data, a model space and a learning function, and assumes no structure on any of them.

Data. Let $\mathcal{Z}$ be a data space, an arbitrary set. An example $z \in \mathcal{Z}$ may be a labeled pair, an unlabeled input, a token sequence, or one party's block of features. A dataset $D \in \mathcal{Z}^*$ is a finite multiset, and two datasets of one protocol may live in different spaces, as an unlabeled pretraining corpus D_{pre} and a labeled fine-tuning set D_j do. Client $j \in \{1, \dots, N\}$ holds D_j of size n_j. The partition is horizontal when each client holds whole examples, and vertical when every example is split into blocks and client j holds one block of each, as in vertical and split learning.

Models. Let $\mathcal{M}$ be a model space, an arbitrary set. A model $m \in \mathcal{M}$ is used through an evaluation map $\mathsf{ev} : \mathcal{M} \times \mathcal{U} \to \mathcal{O}$ from a model and a query to an output, which may be a score vector, a label, a loss, a generated sequence or an embedding. A hypothesis class is a subset $\mathcal{H} \subseteq \mathcal{M}$. A parameterized class $\theta \mapsto m_\theta$ is one instance and not an assumption.

Learning function. A learning function is a randomized map

$$\mathsf{L} : \mathcal{M} \times \mathcal{Z}^* \times 2^{\mathcal{M}} \to \mathcal{M}, \qquad (m, D, \mathcal{H}) \mapsto \mathsf{L}(m, D, \mathcal{H}),$$

from a starting model, a dataset and a hypothesis class to a model, which lies in $\mathcal{H}$ for every learning function in the placed papers. The loss, the optimizer and the number of steps are internal to L. Pretraining is $\mathsf{L}(m_{\text{init}}, D_{\text{pre}}, \mathcal{M})$. Training with frozen parts passes the class of models that agree with m outside the trained parameters, so LoRA and head training differ from full fine-tuning only in $\mathcal{H}$. Aggregation is a learning function whose data space is $\mathcal{M}$ itself: its dataset is the multiset of client models, FedAvg is their mean, a robust aggregator is a robust estimator of location, and HE-OFT's merge is a weighted mean. When $\mathcal{M}$ is a vector space the update is $\mathsf{U}(m, D, \mathcal{H}) = \mathsf{L}(m, D, \mathcal{H}) - m$, and for one gradient step it is the derivative of the dataset's loss evaluated at m.

A protocol is a finite directed acyclic graph. Each node applies a learning function or an evaluation map to arguments $(m, D, \mathcal{H})$ that come from datasets, public objects or earlier nodes, and it emits an output. Every node value is a function of the datasets, the honest randomness and the adversary's choices.

The adversary

Each node value reaches the adversary through a channel $\kappa_v \in \{\text{id}, \ \Sigma, \ \mathsf{Enc}, \ \mathcal{N}_\sigma, \ \perp\}$: the plaintext value, a sum over a set of clients, a ciphertext under a key the adversary does not hold, added noise, or nothing. A decryption policy releases $g(v)$ for some functions g of encrypted values, and for approximate HE the released value carries the decryption noise as well. The view is

$$\mathsf{View} = \big(\kappa_v(v)\big)_{v \in S} \text{ together with oracle access to } \mathsf{ev}(m^*, \cdot),$$

for the set S of node values it sees and the served model m^*. The evaluation map of the served model fixes what an answer reveals.

The choices C are the arguments and outputs of nodes that the adversary sets.

- A data argument: its own dataset, or poisoned examples.
- A model argument: the model broadcast to clients, or a different one for each client.
- A class argument: the architecture, or an inserted module.
- The learning function of a node: training code shipped to clients, the loss that trains a client's layers, or the aggregation rule and schedule.
- The output of a node: an arbitrary update, a returned gradient, or a ciphertext presented for decryption.
- Queries u to the served model.

A semi-honest adversary chooses queries only. Every malicious model in the placed papers chooses more.

The goal is one of three games, each with a baseline, the best value reached from public knowledge alone.

- Distinguish. Two hypotheses H_0, H_1 on $(D_1, \ldots, D_N)$. The discriminator outputs a bit, and $\mathsf{Adv} = \Pr[1 \mid H_1] - \Pr[1 \mid H_0]$. Membership is a pair that differs in one record of D_j. Property inference is a pair of distributions for D_j. Telling a pretraining member from a fine-tuning member is $z \in D_{\text{pre}}$ against $z \in D_j$. HE-OFT's content game is this goal with $D_j \in \{D_0, D_1\}$ of equal size.
- Invert. A target τ of the datasets or the models, and a tolerance η. The adversary wins when $d(\hat{\tau}, \tau) \le \eta$. Reconstruction takes τ a record of D_j, and model extraction takes τ a model that agrees with m^* on queries, scored by fidelity.
- Steer. A target set $T \subseteq \mathcal{M}$. The adversary wins when the trained model lands in T under its choices, while a defender's discriminator between honest and adversarial outputs has advantage at most δ. Untargeted poisoning takes T the models of high expected loss, and a backdoor takes T the models whose outputs on triggered queries satisfy a predicate the adversary fixes, such as a target label or a target completion.

The stealth constraint on steering is itself a distinguish game, played by the defender. A client that tests whether the server deviated plays the same game.

The order, and the two kinds of result

Write $(\mathsf{View}_1, C_1) \preceq (\mathsf{View}_2, C_2)$ when $C_1 \subseteq C_2$ and $\mathsf{View}_1 = g(\mathsf{View}_2, r)$ for a randomized map g with independent randomness r. Proven, by the data-processing inequality for distinguishing and by simulation for inversion: $\preceq$ implies that the first adversary's best distinguishing advantage and best inversion success are at most the second's. The best distinguishing advantage equals the total variation distance between the laws of View under H_0 and H_1 (Neyman and Pearson). For efficient adversaries, g must be efficient.

A *collapse* at a goal shows that the weaker adversary attains the stronger one's best value. For distinguishing this holds exactly when View_1 is a sufficient statistic of View_2 for (H_0, H_1). A *separation* is a dataset law, learning function and hypothesis class with a large gap. Instances, as the sweeps report them unless marked:

- Labels against the head, distinguish. Ordering proven in HE-OFT, read in full: $Q_{\text{tot}} = (t-1)Q$ label answers give a deviating coalition no more than both shared heads in plaintext, and the white-box advantage measures near zero (KSK26).
- Loss against parameters, distinguish. Collapse under a Gibbs law on the parameters (Sablayrolles et al., ICML 2019, from memory). Separation claimed for SGD (Suri, Zhang and Evans, TMLR 2024, outside the venue rule).
- Each client's update against the sum, invert. Collapse when the server chooses a different model argument for each client, since Σ then equals one update (Pasquini, Francati and Ateniese, CCS 2022).
- Honest against chosen model argument, invert. Separation by construction, proven under an RIP condition (ARES, Gong et al., S&P 2026) and empirical (SEER, Garov et al., ICLR 2024, and LOKI, Zhao et al., S&P 2024).
- Chosen model argument through a local DP channel, distinguish. Proven lower bounds for every LDP mechanism (Nguyen et al., ICML 2025).

- Steer with one's dataset against steer with one's update. Proven separation in stability for convex losses (Boudou et al., ICML 2026).
- Steer against the defender's discriminator. Proven for distance-based aggregators that perfect detection implies no learning (Raynal and Troncoso, S&P 2025).

HE-OFT on the interface

Pretraining on D_{pre} gives the backbone φ, public and frozen. Client j runs $\mathsf{L}(m_0, D_j, \mathcal{H}_j)$, where m_0 is the backbone with the shared adapter initialization and the public head initializer θ_0, and $\mathcal{H}_j$ holds the backbone and the adapter's down-projection fixed. The output is the adapter A_j, kept local, and the head h_j. The node emits $\mathsf{Enc}(\mathrm{diag}(g_j)\Delta_j)$ and $\mathsf{Enc}(g_j)$, with $\Delta_j = h_j - \theta_0$ and g_j the per-class counts. The aggregation node is a learning function on the heads, the coverage-weighted mean, whose row c is θ_0 plus $\sum_j n_{j,c}\Delta_{j,c} / \sum_j n_{j,c}$, evaluated under encryption. The served model is the shared head θ^* on the querier's own features, and its evaluation map returns the label alone. A quorum of t clients key-switches each label to the querier, up to Q times per client.

- Semi-honest server with $t-1$ clients. The view is ciphertexts, the public counts n_j, the selected index and the coalition's own answers. Proven simulation security under IND-CPA and a smudging hypothesis. The paper reports that its measured configuration does not meet that hypothesis and recommends 2^{25} smudging with a duplicate-query check (KSK26).
- Deviating clients, honest server. The choices are node outputs (uploads, shares) and queries, including arbitrary feature vectors. The distinguish goal is bounded by a negligible term plus what Q_{tot} label queries give, under three readings on deviating messages that HE-OFT does not enforce by proofs. No steering claim is made. Proven, from the aggregation formula: since no check ties g_j to n_j or to the data, a client that reports a large $g_{j,c}$ moves row c of θ^* as close to its own $\Delta_{j,c}$ as it likes.
- Deviating server. Its choices include the ciphertext presented to the quorum, a node output. HE-OFT proves that a client's decision to contribute a share cannot depend on the plaintext, so no protocol in which clients decrypt what the server presents admits a deviating server, and at four classes one request carries the whole head (KSK26). The paper names a checker that recomputes the deterministic serving circuit, or verifiable HE, as the remedy.

FedAvg, split and vertical learning on the interface

In FedAvg each client node runs K local SGD steps, $\mathsf{L}(m_t, D_j, \mathcal{M})$, and the server sees each update $\mathsf{U}(m_t, D_j, \mathcal{M})$, or only their sum under secure aggregation. Gradient inversion is the invert game on $D_j \mapsto \mathsf{U}(m_t, D_j, \mathcal{M})$. A malicious server sets the model and class arguments so that this map is injective with a computable inverse (ARES and LOKI), or sets a different model argument per client to remove Σ (Pasquini, Francati and Ateniese). With LoRA the class argument confines the update to the adapter, and the update stays invertible while the token count is below the embedding dimension (DAGER, Petrov et al., NeurIPS 2024). In split learning the view is the client's features and their gradient, and a malicious server sets the learning function that trains the client's layers (Pasquini, Ateniese and Bernaschi, CCS 2021). In vertical learning the label holder sets the output of its node, the gradients it returns, and the URVFL attack uses this to invert the other parties' features (Yao et al., NDSS 2025).